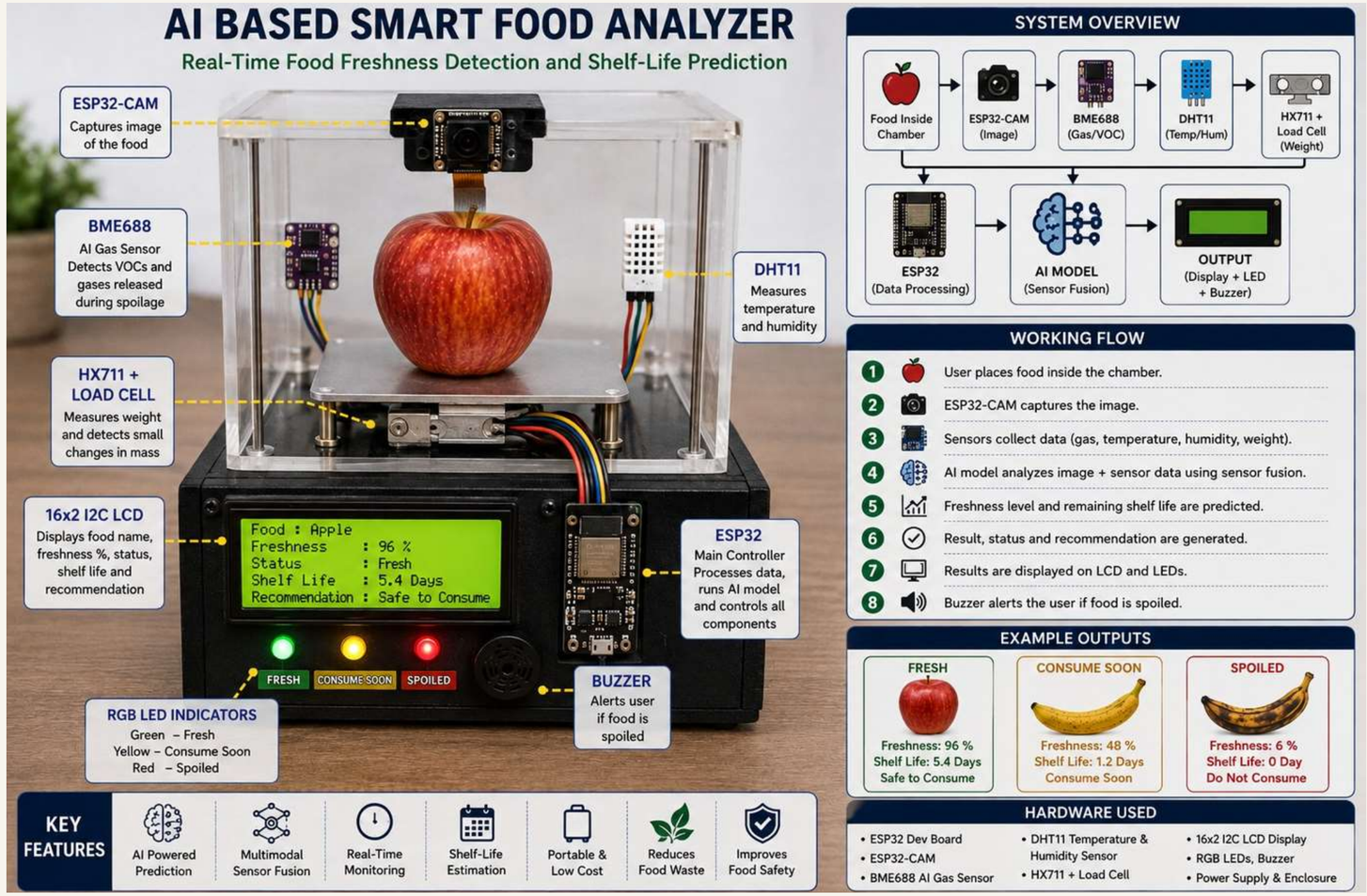
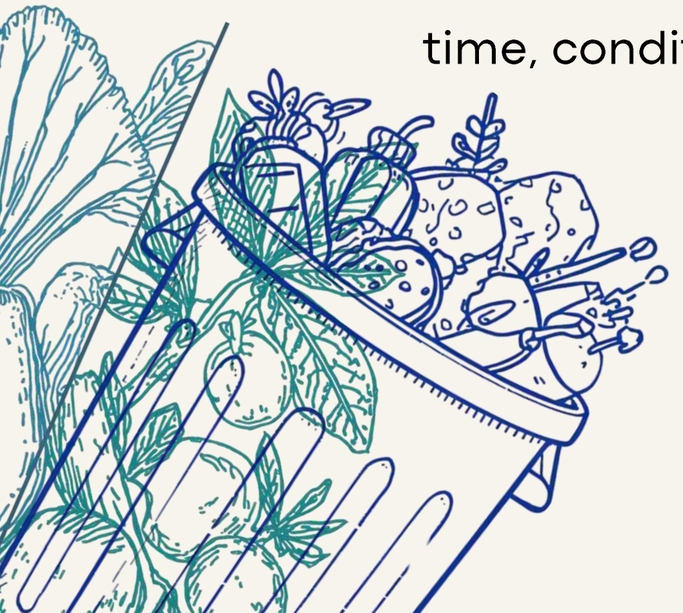
AI-Based Smart Food Analyzer



Problem Statement

Expiry dates are printed at the time of manufacturing and assume ideal storage conditions — but real-world storage rarely meets those conditions. Temperature fluctuations, humidity, improper handling, and power outages can cause food to spoil well before the printed date, or remain safe beyond it.

Every year, millions of people suffer from foodborne illnesses caused by consuming food that appeared "safe" by label. Simultaneously, edible food is discarded due to overly cautious date labels. There is a critical need for real-time, condition-based food freshness detection — not just static date stamps.





Social Relevance

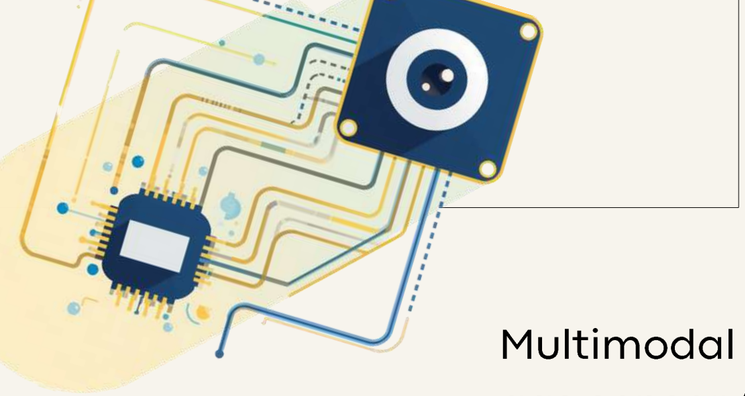
🌍 The Food Waste Crisis

1.3 billion tonnes of food is wasted globally every year (FAO). In India alone, food worth ₹92,000 crore is wasted annually. A significant portion of this waste occurs due to inability to assess real-time freshness — consumers discard food that is still safe, or worse, consume food that has already spoiled.

🛡️ The Food Safety Emergency

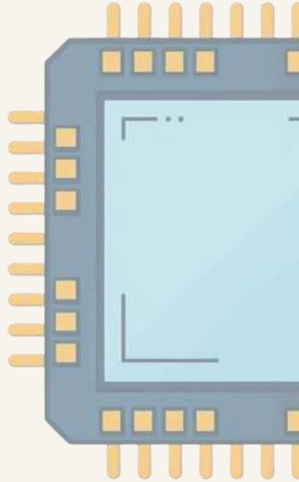
Over 600 million people fall ill each year from consuming contaminated food (WHO). Spoiled food causes foodborne diseases like salmonellosis and food poisoning, placing immense pressure on healthcare systems. Vulnerable groups — children, elderly, and hospital patients — are at the highest risk. A smart, real-time freshness detection system directly addresses both food waste reduction and food safety, making it a critical social innovation.

Innovation

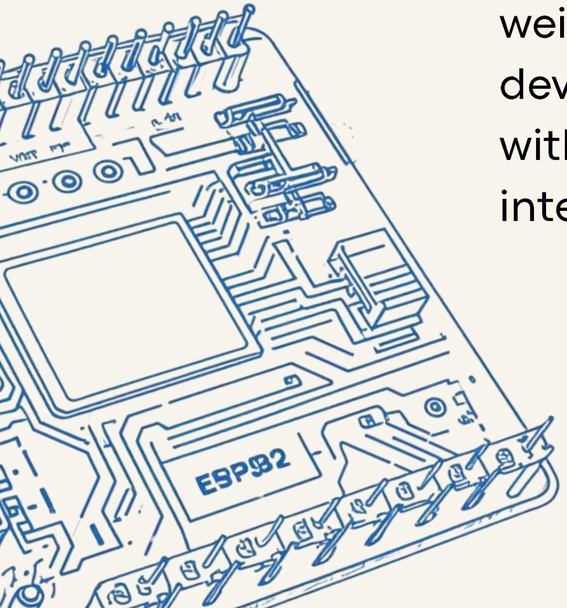




Multimodal AI Food Analyzer: Our innovation combines a camera module, gas sensor (BME688), temperature & humidity sensor (DHT11), and load cell — all fused through an ESP32 microcontroller. A lightweight CNN model analyzes visual features (color, texture, mold) in real time, while sensor data provides environmental and chemical context for a comprehensive freshness verdict.



Unlike single-sensor systems, this multimodal approach cross-validates signals: visual degradation is confirmed by gas composition changes and weight loss, dramatically reducing false positives. The AI model runs on-device, delivering instant freshness scores — Fresh, Caution, or Spoiled — with LED, buzzer, and LCD alerts. No cloud dependency. No delay. Just intelligent, real-time food safety.





What Makes It Different

Single-Sensor Limitations

Traditional food freshness detection relies on a single data source — either a visual camera, a gas sensor, or a weight scale alone. These methods are highly prone to false positives and missed detections. A camera cannot detect invisible spoilage gases; a gas sensor cannot identify mold or discoloration. Single-sensor systems fail in real-world, multi-variable environments.

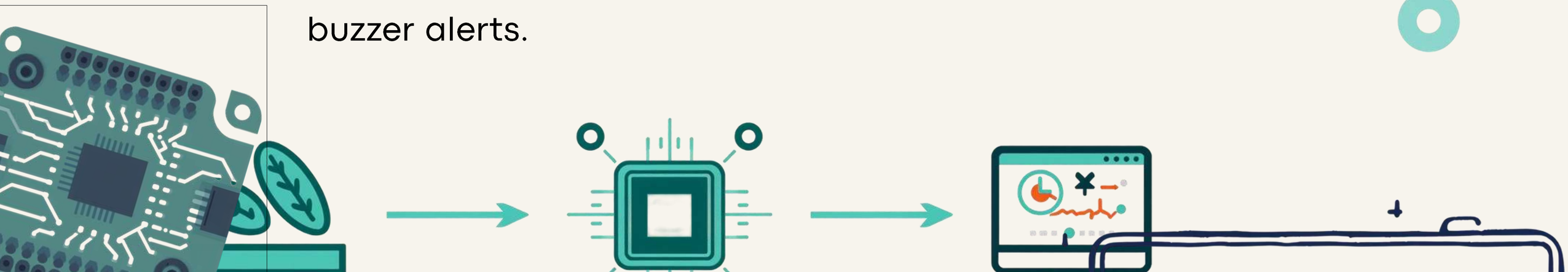
Our Sensor Fusion Advantage

innovation fuses four independent data streams: camera (visual spoilage), BME688 (VOC gas & odor), DHT11 (temperature & humidity), and load cell (weight loss). The ESP32 microcontroller aggregates all signals through an AI model, delivering a holistic, multi-dimensional freshness verdict — far more accurate and reliable than any single-sensor approach.

Working Principle

Step 1–3: Food is placed on the load cell (weight captured) → Camera module captures a high-resolution image → BME688 gas sensor and DHT11 measure volatile compounds, temperature & humidity in real time.

Step 4–6: ESP32 microcontroller fuses all sensor data → On-device AI model processes image + gas + environmental inputs → Freshness score is classified (Fresh / Moderate / Spoiled) and displayed on LCD with LED & buzzer alerts.



System Architecture

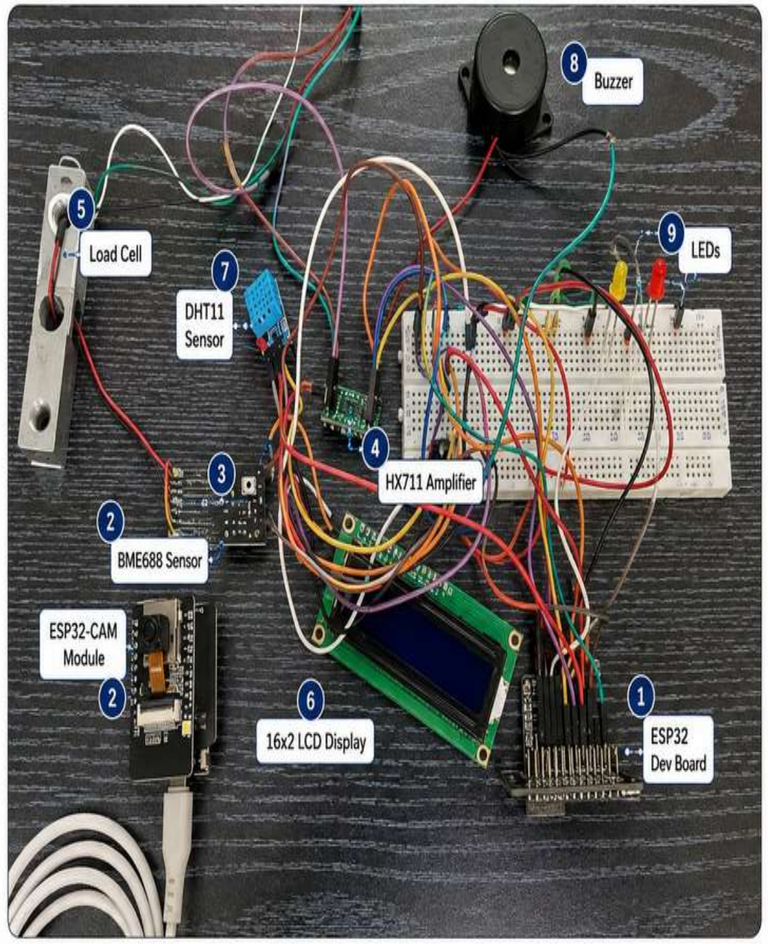
Input Sensors

📷 Camera Module
Captures real-time food images for visual AI analysis.
🌡️ BME688 Sensor
Measures gas, humidity, pressure & temperature.
💧 DHT11 Sensor
Monitors ambient temperature and humidity levels.
⚖️ Load Cell
Detects weight changes indicating spoilage or gas release.

Processing & Output

🟢 ESP32 Microcontroller
Central processing unit running AI inference and sensor fusion.
📱 LCD Display
Shows real-time freshness status and sensor readings.
💡 LEDs (Red/Green)
Visual alerts — Green: Fresh, Red: Spoiled or Warning.
🔔 Buzzer
Audible alert triggered when food is detected as unsafe.

COMPLETED CIRCUIT – COMPONENT LIST



No.	Component	Description	Qty
1	ESP32 Dev Board	Main controller (Wi-Fi, processing, system control)	1
2	ESP32-CAM Module	Captures images for crop monitoring and disease detection	1
3	BME688 Sensor Module	Measures temperature, humidity, pressure, gas (environmental monitoring)	1
4	HX711 Load Cell Amplifier	Amplifies load cell signal for weight measurement	1
5	Load Cell (Weighing Sensor)	Measures weight of produce / crop yield	1
6	16x2 LCD Display	Displays sensor data, status, and messages	1
7	DHT11 Temperature & Humidity Sensor	Measures temperature and humidity	1
8	Buzzer	Provides alert sound and notifications	1
9	LEDs (Red & Yellow)	Visual indicators for status, alerts	2
10	Jumper Wires & Breadboard	Connections between components for prototyping	Many

Note: All components are connected together for environmental monitoring, image capture, and weight measurement with user feedback (LCD, LEDs, Buzzer).

Actual Components

Hardware Components Used in the Prototype:

- **ESP32 Microcontroller** – Central processing & Wi-Fi/Bluetooth connectivity
- **BME688 Gas Sensor** – Detects VOCs, CO2, humidity & temperature
- **DHT11 Sensor** – Ambient temperature & humidity monitoring
- **Load Cell + HX711** – Measures food weight for freshness tracking

Additional Components:

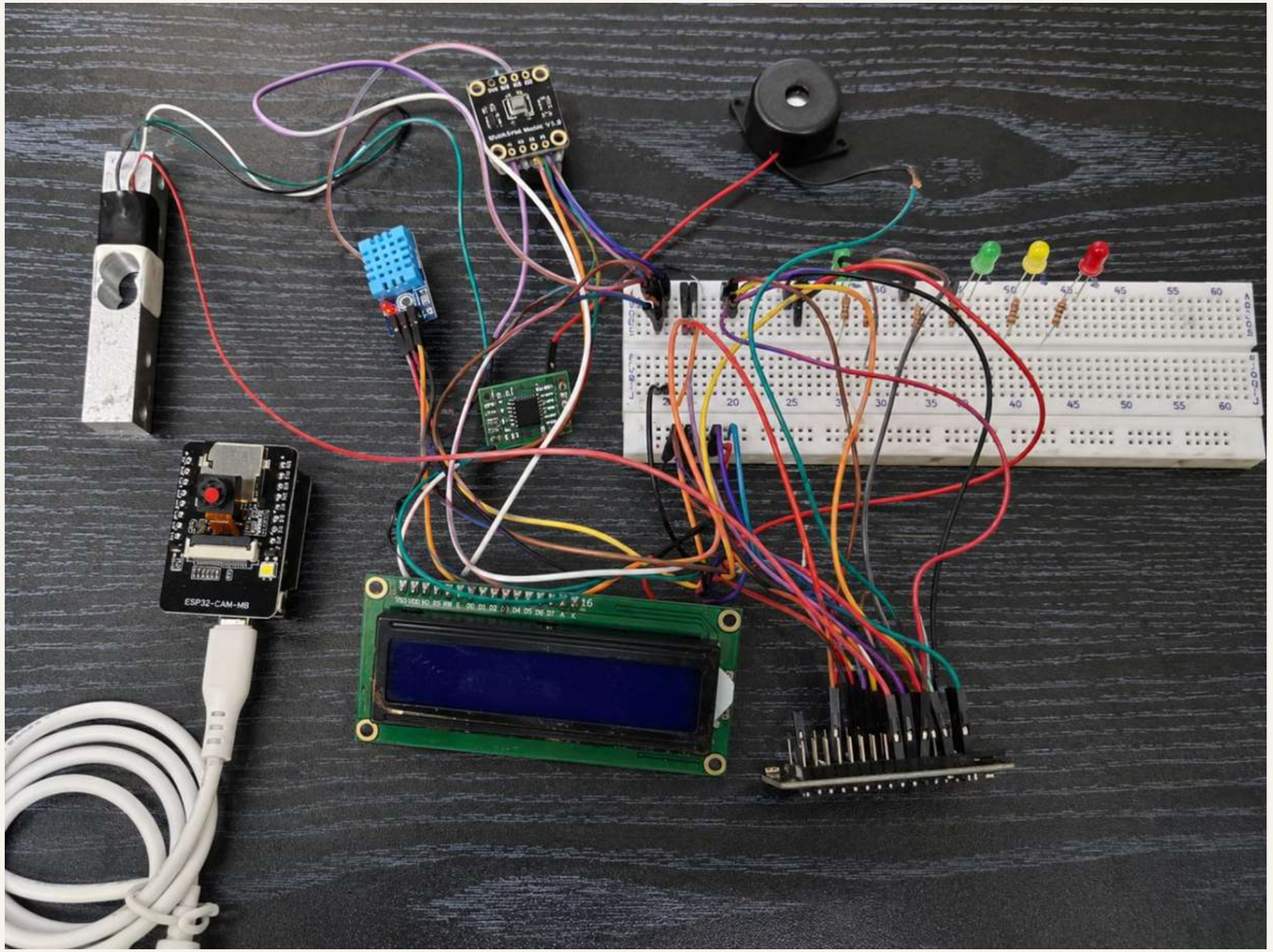
- **Camera Module (OV2640)** – Captures real-time food images for AI analysis
- **16x2 LCD Display** – Shows freshness status and sensor readings
- **RGB LEDs** – Visual freshness indicators (Green / Yellow / Red)
- **Buzzer** – Audio alert for spoiled or unsafe food detection
- **Breadboard, jumper wires, resistors, capacitors & 5V power supply**



Actual Prototype

The assembled Smart Food Analyzer prototype integrates all hardware components on a breadboard/PCB. The ESP32 microcontroller serves as the central processing unit, interfacing with the BME688 gas sensor, DHT11 temperature & humidity sensor, and load cell — all connected via GPIO pins and I2C/SPI protocols.

The OV2640 camera module captures real-time food images for AI inference. An LCD display shows freshness status, while RGB LEDs and a buzzer provide instant visual and audio alerts. The compact wiring layout ensures reliable sensor readings and low-latency AI-based freshness detection output.



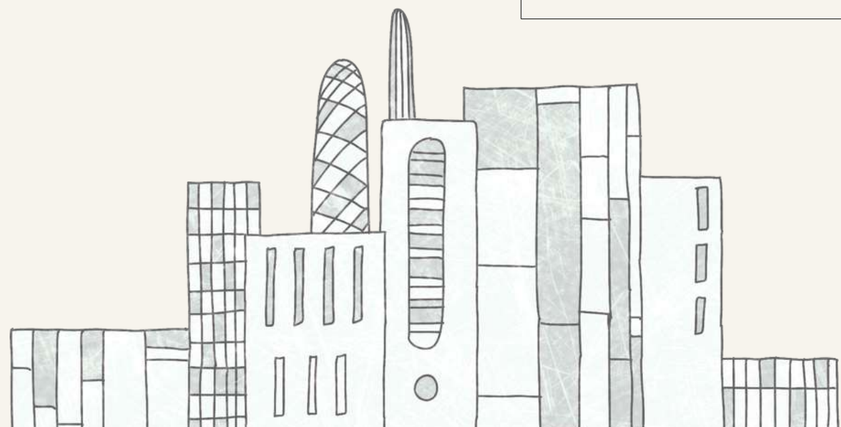
Prototype Concept / Enclosure

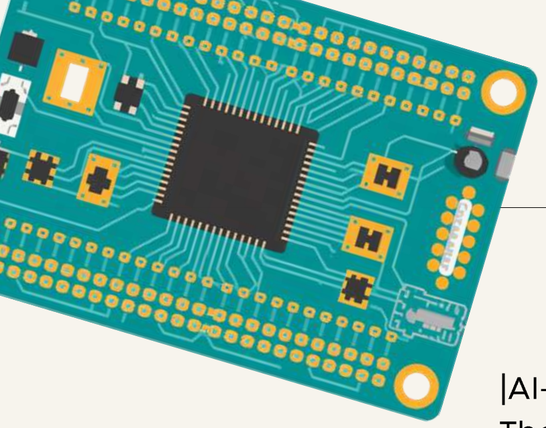
The Smart Food Analyzer is housed in a compact, user-friendly enclosure designed for countertop use. The front panel features an LCD display showing real-time freshness status, LED indicators (green/yellow/red), and a buzzer for audio alerts. The top surface includes a food placement tray integrated with the load cell sensor for weight measurement.

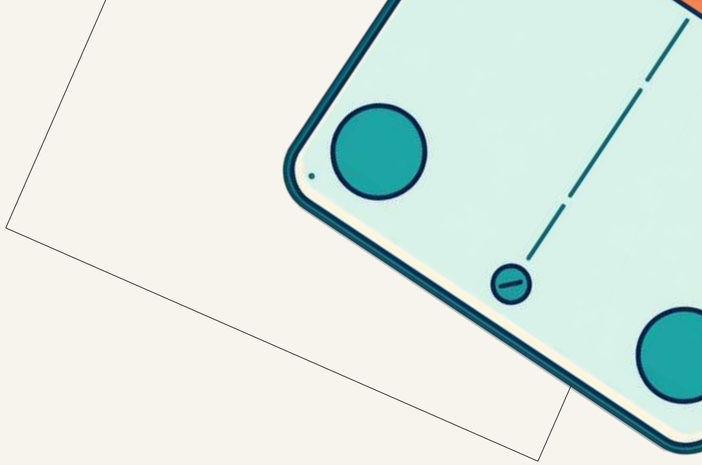
The side vents house the BME688 gas sensor and DHT11 for ambient air sampling. A wide-angle camera module is embedded at the front for visual food analysis. The ESP32 microcontroller and power module are neatly enclosed at the rear. The design prioritizes portability, hygiene, and ease of use for homes, kitchens, and retail environments.











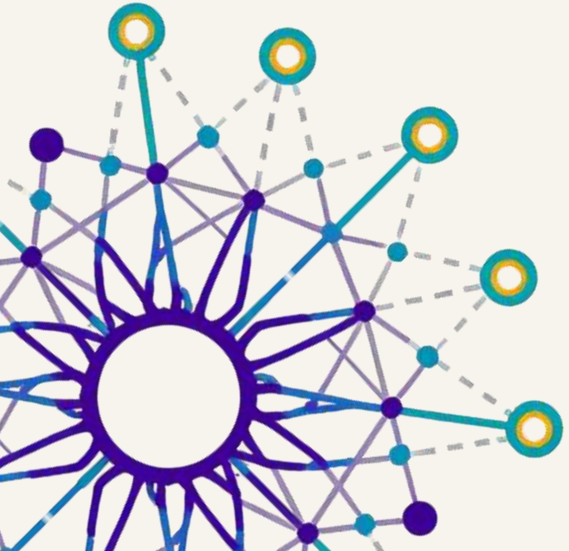
AI & Sensor Fusion

|AI-Based Smart Food Analyser

The system fuses data from five modalities: a camera captures visual texture and colour changes; the BME688 gas sensor detects volatile organic compounds (VOCs) released during spoilage; DHT11 measures ambient temperature and humidity; and a load cell tracks mass loss from moisture evaporation. Each stream is pre-processed independently before being passed to the AI fusion engine.

The ESP32 microcontroller runs a lightweight multimodal inference model that assigns weighted confidence scores to each sensor input. A decision-fusion algorithm combines these scores to classify freshness as Fresh, Borderline, or Spoiled. This multi-parameter approach eliminates single-sensor blind spots — for example, visually intact food that emits spoilage gases is correctly flagged, ensuring accurate and reliable real-time food safety assessment.









Output & User Interface

Freshness States

The system classifies food into four freshness levels: FRESH (green LED, LCD displays "Safe to Consume"), MODERATE (yellow LED, advisory message), STALE (orange LED, warning alert), and SPOILED (red LED + buzzer alarm, "Do Not Consume" alert). Each state is determined by fusing image analysis, gas sensor readings, temperature, humidity, and weight data in real time.

Alert System

Multi-modal alerts are triggered instantly: visual LED indicators (green/yellow/orange/red), an LCD 16x2 display showing freshness score and status message, and a buzzer alarm for spoiled food. Results are also transmitted via ESP32 Wi-Fi to a mobile dashboard, enabling remote monitoring and logging of food freshness history with timestamps.









Applications

Everyday Use

🏠 Homes: Detect spoiled fruits, vegetables, and dairy before consumption. Reduce household food waste and prevent foodborne illness.
🏪 Retail Stores: Real-time shelf monitoring to flag near-expired or spoiled stock. Automate freshness-based discounting and restocking alerts.

Institutional Use

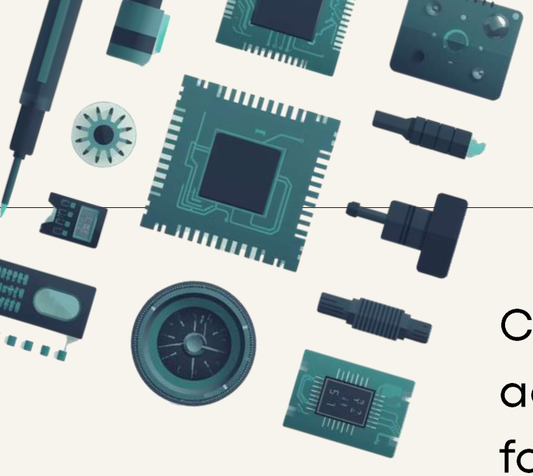
Restaurants & Canteens: Ensure ingredient freshness before cooking. Maintain food safety compliance and reduce liability from spoiled meals.
🏥 Hospitals & Care Facilities: Critical freshness verification for patient meals. Zero-tolerance food safety in immunocompromised environments.











Limitations & Future Scope

Current Limitations: The prototype requires controlled lighting for accurate image analysis and may struggle with packaged or opaque food items. Sensor calibration drifts over time, requiring periodic recalibration. The AI model is trained on limited food categories and may not generalize to all regional food types. Real-time processing is constrained by ESP32 compute capacity, and the current enclosure is not yet waterproof or food-grade certified for commercial use.

Future Scope: Expand the AI model with a larger, diverse dataset covering 100+ food categories using cloud-based training. Integrate IoT connectivity for remote monitoring via mobile app with push alerts. Develop a miniaturized, food-safe enclosure for smart refrigerator integration. Incorporate e-nose sensors for enhanced volatile compound detection. Explore blockchain-based food traceability and deploy edge-AI chips for faster, offline inference in commercial supply chains.

Conclusion

The AI-Based Smart Food Analyzer represents a meaningful leap forward in how we detect and respond to food spoilage. By combining real-time sensor data — gas, temperature, humidity, and weight — with AI-powered image analysis, the system delivers condition-based freshness monitoring that goes far beyond the limitations of printed expiry dates.

- This innovation addresses critical challenges in food safety and waste reduction across homes, stores, restaurants, and hospitals.
- Real-time alerts, an intuitive interface, and multimodal sensor fusion make it a practical, scalable, and impactful solution.
- — Smart Food Analyzer: Ensuring every bite is safe, fresh, and informed by data.







MADE BY: SHAMS TABREJ